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Collective behaviour Seminar Paper: Sheep herding behaviour

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Sheep are well known herding animals, and we aim to simulate their collective behaviour. In particular, we want to focus on the main instincts of grazing and escaping a threat. When grazing, the group spreads out to maximize area available for each individual. The escaping response is triggered when the sheep are startled by an approaching threat. The group flocks and flees together(1). This fleeing behaviour follows a general zig-zag pattern, with a herding dog then adapting to the sheep's movements. In this project, we attempt to reproduce these different behaviours.

Collective behaviour | Sheep | Herd | Simulation

Many animals display collective behaviour. Among these are bird swarms, fish shoals, and animal flocks. These behaviours often emerge when there is benefit in staying in dense formations to escape and confuse predators. This also applies to sheep, a much studied species, that has been domesticated by humans for millennia. In this work we will go through the observed specificities of sheep collective behaviour and attempt to model it.

After this introduction, we will review the related work on our topic, along with the proposed modelings. Following, we will described the implementation of our own modeling.

1. Related Work

This section presents an overview of existing work regarding sheep flock behaviour, as well as a review of three proposed agent-based models. First, the general observations of behaviour will be discussed, before different propositions of implementations from these works are briefly presented.

1.1. General observations. In a 2024 paper, Jadhav et al. investigate the patterns sheep and herding dogs follow when interacting, specifically when the sheep are actively being herded. To examine the behaviour, they set up an experiment, where a flock of sheep and a herding dog were both equipped with a GPS tracker. Then the dog, on a shepherd's instructions, herded the flock between two reference points. The movement of the herd propagates from the front to the back of the flock, meaning the leading sheep has the highest influence on direction. The sheep's relative positions to each other do not change during the chase: the leaders remain the same. The herding dog, at the back of the herd, moves in a zig-zag pattern around the herd to influence its direction. This is pictured in Figure 1.

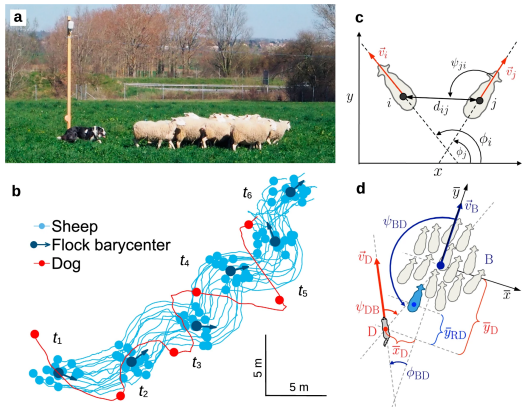


Fig. 1. The patterns of fleeing sheep, picture from Jadhav et al. (2)

When sheep graze, they tend to spread into smaller groups. Then, if a perceived threat approaches, a "fast packing event" (1) occurs

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to avoid individual predation, triggered by an individual's startle response. The sheep rapidly flock together and flee from the perceived threat once it approaches. This is again replaced by grazing when the herd has settled down. Ginelli et al. describe this as "intermittent collective dynamics with large density oscillations" (1).

The observations of small sheep groups (N = 2, 3, 4) by Gómez-Nava et al. align with this observation of grazing periods followed by sudden movement in a random direction, not affected by previous startle events. The transitions between grazing and movement is here, too, observed to be sudden and rapid, with a random leader selection (3). This aligns with Jadhav et al.'s observation that there are no fixed leader-follower pairs in the sheep flock (2).

An additional factor in high-threat scenario positioning within the herd is "selfish-herd theory". In a 2012 experiment, King et al. equipped a herd of 46 sheep with GPS trackers. Although the initial distance between sheep varied for each observed instance, in a threat scenario, they found the sheep tended to gravitate toward the center of the herd until a tight cluster was formed (4). This confirms W.D. Hamilton's claim that the imperative to shield oneself is a prominent part of herding behaviour (5).

1.2. Mathematical equations for agent models.

1.2.1. Jadhav et al.'s modeling for escaping simulation. Based on their observations, Jadhav et al. proposed an agent model where the movement of an individual is determined by a sum of vectors. The deplacement of a sheep is ruled by the following equation:

$$\vec{r}_i^{n+1} = \vec{r}_i^{n} + l\vec{e}(\phi_i^{n+1})$$

where $\vec{r}_i^{n+1}$ is the new position of the sheep, $\vec{r}_i^{n}$ its previous position, l the distance traveled during the elapsed time Δt, $\vec{e}$ a unit vector and ϕ_i^{n+1} the sheep's angle determined by:

$$\phi_i^{n+1} = \alpha\vec{e}(\phi_i^{n}) + \vec{S}_{\text{social}}^{i,n} + \vec{R}_{\text{Dog}}^{i,n} + \epsilon\vec{\mathcal{N}}_i^{n},$$

were $\vec{\mathcal{N}}_i^{n}$, is a random vector. This structure takes into account both the "social vectors", coming from the interaction with neighbouring sheep:

$$\vec{S}_{\text{Att}}^{i} = \frac{w_{\text{Att}}}{n_{\text{Att}}}\sum_{j=1}^{n_{\text{Att}}}\frac{\vec{r}_j - \vec{r}_i}{\|\vec{r}_j - \vec{r}_i\|}, \vec{S}_{\text{Ali}}^{i} = \frac{w_{\text{Ali}}}{n_{\text{Ali}}}\sum_{j=1}^{n_{\text{Ali}}}\vec{e}_j, \vec{S}_{\text{Rep}}^{i} = -\frac{w_{\text{Rep}}}{n_{\text{Rep}}}\sum_{j=1}^{n_{\text{Rep}}}\frac{\vec{r}_j - \vec{r}_i}{\|\vec{r}_j - \vec{r}_i\|}$$

were Att stands for attraction, Ali for alignment and Rep for repulsion, as well as the repulsion vector from the dog:

$$\vec{R}_{\text{Dog}}^{i} = -w_{\text{Dog}}\frac{\vec{r}_{\text{Dog}} - \vec{r}_i}{\|\vec{r}_{\text{Dog}} - \vec{r}_i\|},$$

The dogs position is calculated in relation to the center of the flock, or to a stray sheep, should that event occur. The dog will also adjust its position in short intervals:

Thus, the turning angle for the flock is dependent on the dogs position, with the flock turning away from the dog as it moves towards the flanks. The sheep are also modeled to avoid collisions with each other. Because the dog tends to move in a zig-zag-pattern, the changing flock angle keeps the sheep in a relatively straight line and allows the dog to guide the flock effectively.

This model has the advantage of accounting for the possibility of a single sheep getting lost. It also not only models a herding dog, but its modeling is relatively simple as it only takes into account the flocks position and the dog's herding goal. However, the herding period and subsequent interactions between sheep of a flock are not taken into account. Regardless, this method is efficient, seemingly accurate, and suitable for expansion. (2)

1.2.2. Gómez-Nava et al.'s modeling for behavioural shifts. Gómez-Nava et al. studied behavioural shifts by tracking the position of 4 Merino ewes. They observed a pattern where the herd displays regular shifts between collective motion phases

(CMP) and grazing phases (GP). As they were unable to find an agent-based model that accurately represented the CMP, Gómez et al. proposed a model where

$$\dot{\mathbf{x}}_i(t) \;=\; v(t)\,\widehat{\mathbf{e}}(\theta_i)$$

denotes the direction and speed of the individual and

$$\dot{\theta}_i(t) \;=\; \sum_{j=1}^{N} A_{ij}(t)\, g_{ij}(\mathbf{x}_i, \theta_i, \mathbf{x}_j, \theta_j) \;+\; \sqrt{2\,D_\theta}\xi_i(t),$$

describes the transition between GP and CMP. Here, A_{ij} denotes the influence of individual i on individual j, which is 0 in GP, and $\xi_i(t)$ represents a random noise factor. Following these equations, sheep are moving independently outside of specifically the CMPs. The group remains cohesive by using the hierarchical following mapped by the variable A_{ij} (3), similar to the previously discussed model. This model takes into account both GP and CMP, and adapts to different hierarchy, depending on the position inside the herd. However, the authors do not provide a critical evaluation of their own works when deployed and used a very small initial sample size of 4. These points of criticism are offset by the model's great performance in regards to flock cohesion and leader-follower representation. Unfortunately, as opposed to the first model investigated, this one only accounts for the sheep's behaviour and not the dog's.

1.2.3. Ginelli et al.'s modeling for conflicting imperatives. As opposed to the other two groups' relatively small sample size, Ginelli et al. observed a group of 100 ewes in their study. They tracked their movements over the course of 5 1-hour intervals, and found the same alternating grazing and movement shift as Gómez-Nava et al. Their modeling aims to use group attraction to show these "avalanche dynamics" of fast packing events. They use three formulae for the sheep movement, one each for each behaviour state q, where $q(0)$ represents stationary, $q(1)$ walking, and $q(2)$ running sheep.

$$\mathbf{r}_i^{t+\Delta t} = \mathbf{r}_i^t + \Delta t\, v(q_i^t)\mathbf{s}_i^{t+\Delta t},$$

$$\theta_i^{t+\Delta t} = \mathrm{Arg}\left[\sum_{j\in\mathcal{M}_i} \mathbf{s}_j^t\right] + \psi_i^t \quad (\text{if } q_i^t = 1),$$

$$\theta_i^{t+\Delta t} = \mathrm{Arg}\sum_{j\in\mathcal{V}_i}\left[\delta_{2,q_j^t}\mathbf{s}_j^t + \beta f(r_{ij}^t)\mathbf{e}_{ij}^t\right] \quad (\text{if } q_i^t = 2)$$

Here, s_i^t is the individual's direction, Δt is the time step, and $v(q)$ represents the sheep's speed, dependent on its state. Walking sheep try to align themselves with their neighbours, all sheep within an interaction range. This results in smaller grazing subgroups. Running sheep, in contrast, align themselves with their closest Voronoi neighbours regardless of distance. This results in a flocking pattern. Transition rates between walking and running, and running and stopping are both dependent on the number of neighbours performing the same operation. (1)

This modeling was founded on larger samples than the other two, but this time via using transitions between different states. However, this modeling does not takes into account the threat aspect of the flocking action but instead use probability determined transitions.

2. Implementation

In this section, the specific implementation chosen will be discussed.

Our goal is to combine the previously described agent-based models in a more complete model. imilarly to Gómez et al. and Ginelli et al., we aim to not only represent the fleeing patterns of sheep, but display a more dynamic type of behaviour, where the herd can spread to graze and rest outside of startle responses, and then gather and flee reacting to a threat. The modeling

of the herding dog would allow us to reproduce the conditions of Jadhav et al.'s experiment. (3).

2.1. Language and initial choices. Java was chosen as the language due to its object oriented nature. Every simulated entity, may it be a sheep, a dog or a predator, is associated with a 2D shape, a set of behaviours and a set of self characteristics. For the behaviour part, we will use an approach similar to the one used by Ginelli et al. (1). At each time step, we compute each entity's state, depending on its perceived environment. Then, rather than using a novel approach, we took inspiration from Jadhav et al.'s model, with the entities' movements' resulting from a sum of vectors. However, we used an object-oriented environment to split each vector, computing into a specific "Instinct" class. With this class, each individual's characteristics, such as age, energy, hunger, and fear, can be used in the state change and vector computations, reproducing the diversity of a real herd. These variables are set at the beginning of the simulation. For example, the older sheep would spot the threat later and run slower than younger ones.

Our core program aims to include three states, inspired by Ginelli et al.:

- Grazing, where the individuals remains static
- Foraging, where the individuals walk at a slow pace (Gómez-Nava et al.'s GP)
- Escaping, where the individuals are running to escape a perceived threat (Gómez-Nava et al.'s CMP)

Then once the Escaping is triggered, the corresponding instincts, following Jadhav et al., would be:

- Attraction or Shielding, a vector that pushes an individual toward the centroid of the group as described in selfish herding theory (6)
- Repulsion or Collision avoidance, that pushes an individual opposite to the nearest obstacle
- Alignment or Following, that pushes an individual follow the direction of the herd

Lastly, in the Foraging phase, the Alignment disappears. We aim to use the same set of instincts as with the Escaping, but adapt them to follow a "Maximize area" instinct, pushing each sheep to maximize its grazing space.

This implementation method would then allow us to simulate other states of sheep, such as:

- Scout, watching for eventual threats. For this action, we would temporary extend the vision sight of the individual.
- Sleep, if we add a day cycle to our simulation.

The chasing entities would be modeled with a more straightforward behaviour. For example, for the herding dog, concurrent with Jadhav et al., the modeling would imply two states of one instinct each. Finally, for the display, we use JavaFX for 2D drawings.

2.2. Implementation and validation steps.

1. Implementation of the Java structure.
2. Implementation of a "dummy sheep" with simple instincts, tested by visual display.
3. Implementation of escaping and grazing states, tested by click: the mouse click acts as a threat to test the sheep's reaction.
4. Implementation of the herding dog, tested by tracing the trajectories of the dog and the herd to reproduce the zig-zag observations of Jadhav et al.
5. Further steps could include potential add-ins, like predators, obstacles, visual effects, ...

We already completed steps 1 and 2, and are working on step 3.

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